BUILDING A USB RC METER

Episode 6 Worksheet

Does It Actually Work? Testing, Accuracy and a 2026 Redesign

Work through these after watching the episode. Show your working; answers are on the last page. Values come from the 2014 thesis data unless stated.

Learning objectives checked

- Separate a meter's accuracy from a component's tolerance
- Diagnose error types from data: offset, gain (calibration) and quantisation
- Design a proper validation plan for a measuring instrument
- Specify a modern (2026) version of this meter

Questions

1. My meter read 956.4 Ω; the CD800a read 970 Ω. Compute the % difference and the CD800a's spec window.

2. Stage-1 readings for 1.1 Ω and 9.9 Ω were 4.98 Ω and 14.94 Ω. Express both in counts of 2.49 Ω and estimate the offset.

3. Why is a 260 Ω entry suspicious, and how would you correct its error?

4. Ten readings of a 1.000 kΩ reference give a mean of 0.987 kΩ and σ = 0.6 Ω. Describe the accuracy and precision.

5. Write a datasheet-style accuracy spec if a range shows ±0.8 % of reading plus ±2 counts of 23.2 Ω.

6. Estimate the Stage-1 current change if R_{on} rises by 0.24 %/°C (from 345 Ω) over +20 °C.

7. List four experiments missing from the thesis and what each would prove.

8. Why can't comparing against a ±(1.2 % + 5 dgt) handheld DMM verify a ±0.5 % meter?

Answers

1. (956.4 − 970) / 970 = −1.40 %. Window: 970 ± (11.6 + 5) Ω = 953.4–986.6 Ω, so the difference is inside the reference's uncertainty.
2. 2 counts and 6 counts; expected ≈0.44 and ≈3.98 counts → offset ≈1.6–2 counts (≈4–5 Ω).
3. 260 is not an E24 value; both DMMs read ≈271 Ω, so it's likely 270 Ω: (268.985 − 270) / 270 = −0.38 %.
4. Systematic error −1.3 % (accuracy); precision ≈ ±0.06 % (1σ). Precise but biased → calibrate the gain.
5. ±(0.8 % rdg + 2 counts) on Stage 2, i.e. ±(0.8 % + 46 Ω).
6. ΔR_{on} ≈ 345 × 0.048 ≈ 16.6 Ω; total 507 → 523.6 Ω → current −3.2 %.
7. Supply sweep (ratiometric claim), temperature (D1 / R_{on} drift), repeatability (precision), 1–2 MΩ (upper range claim); also back-to-back capacitor readings and clock verification.
8. The reference's uncertainty (≥1.2 %) is larger than the accuracy being claimed; you need a reference ~4–10× better (TUR ≥ 4:1).

Going further

Sanwa CD800a specifications (Ω ±(1.2 %+5), C ±(5 %+10), 100 µF max): https://overseas.sanwa-meter.co.jp/products/digital_multimeters/cd800a.html

TI CD4051B datasheet (R_{on} vs temperature): https://www.ti.com/lit/ds/symlink/cd4051b.pdf

Cytron UC00C (CH340) price: https://my.cytron.io/p-uc00c-ch340-usb-to-uart-converter

Keysight: four-wire resistance measurement: https://www.keysight.com/us/en/use-cases/measure-resistance-using-four-wire-measurement.html

NIST/SEMATECH e-Handbook of Statistical Methods: measurement process characterization: https://www.itl.nist.gov/div898/handbook/mpc/mpc.htm

AVR Transistortester (open-source component tester): https://www.mikrocontroller.net/articles/AVR_Transistortester